

29/10/18

1. A single plate clutch is required to transmit 8 kW, 1000 rpm. The axial pressure is limited to 70 kN/m². The mean radius is 4.5 times radial width of the friction surface. If both sides of the plates are effective and coefficient of friction is 0.25. Find.

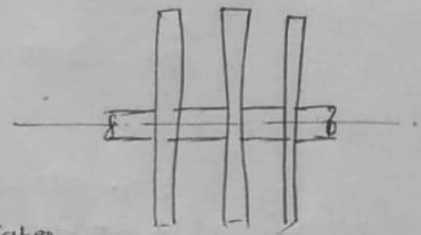
- i) Inner & outer radius of plate, mean radius.
ii) width of friction lin

A.

$$T = \mu F \frac{r_i + r_o}{2} \times n.$$

$$T = \mu F \frac{2}{3} \left(\frac{r_o^3 - r_i^3}{r_o^2 - r_i^2} \right) \times n.$$

no. of plates.



$$Pr = C.$$

$$P_{max} r_{inner} = C.$$

$$\text{mean radius} = \frac{r_o + r_i}{2}$$

$$P = 8 \text{ kW}$$

$$N = 1000 \text{ Rpm.}$$

$$p_i = 70 \text{ kN/m}^2.$$

$$(r_o - r_i) 4.5 = \frac{r_o + r_i}{2}.$$

$$n = 2.$$

$$(r_o - r_i) 90 = r_o + r_i$$

$$P = \frac{2\pi NT}{60}.$$

$$T = \frac{P \times 60}{2\pi N} = \frac{8 \times 10^3 \times 60}{2 \times \pi \times 1000} = \underline{\underline{76.39 \text{ Nm}}}$$

$$T = \mu F \frac{(r_1 + r_0)}{2} \times n. \quad \text{--- (1)}$$

$$F = 2\pi C (r_0 - r_1)$$

$$F = 2\pi \underbrace{P_i r_i}_{\text{max.}} (r_0 - r_1) \Rightarrow \text{substitute (F) in (1)}$$

$$76.3 = 0.25 \times \pi \times 2 \times 70 \times 10^3 \times r_i (r_0 - r_i) \left(\frac{r_1 + r_0}{2} \right) \times 2$$

$$76.3 = 0.25 \times \pi \times 2 \times 70 \times 10^3 \times \frac{89}{91} r_0 \left(r_0 - \frac{89}{91} r_0 \right) \left(\frac{r_0 - r_1}{2} \right) \times 2$$

$$(r_0 - r_1) r_0 = r_0 + r_1$$

$$90 r_0 - 89 r_1 = r_0 + r_1$$

$$89 r_0 - 90 r_1 = 0$$

$$89 r_0 = 90 r_1$$

$$r_1 = \frac{89}{90} r_0$$

$$r_1 = 0.8 r_0$$

$$76.3 = 0.25 \times \pi \times 2 \times 70 \times 10^3 \times 0.8 r_0 (r_0 - 0.8 r_0) \left(\frac{0.8 r_0 + r_0}{2} \right) \times 2$$

$$76.3 = 70000 \pi \times 0.8 r_0 (r_0 - 0.8 r_0)$$

$$= 70000 \pi \times 0.8 r_0 \times 0.2 r_0 \times \frac{0.9}{2} r_0$$

$$76.3 = 70000 \pi \times 0.144 r_0^3$$

$$r_0^3 = 2.399 \times 10^{-3}$$

$$r_0 = 0.1338 \text{ m}$$

$$r_1 = 0.8 r_0$$

$$= 0.8 \times 0.1338$$

$$= 0.1071 \text{ m}$$

2.

A multiplate disc clutch. 55 kW of power at 1800 rpm. Co. efficient of friction for the friction surfaces is 0.1. Axial intensity of pressure is not to exceed 160 kN/m^2 . The internal radius is 80 mm and is 0.75 times external radius find the no. of plates needed to transmit the torque.

A.

Given

$$P = 55 \text{ kW}$$

$$N = 1800 \text{ rpm.}$$

$$\mu = 0.1$$

$$P_i = 160 \text{ kN/m}^2,$$

$$r_i = 80 \text{ mm} = 0.08 \text{ m.}$$

$$r_o = 0.106 \text{ m.}$$

$$\begin{aligned} r_i &= 0.75 \times r_o \\ 80 &= 0.75 r_o \\ r_o &= 0.106 \text{ m} \end{aligned}$$

$$P = \frac{2\pi NT}{60}$$

$$55 \times 10^3 = \frac{2\pi \times 1800 \times T}{60}$$

$$T = \underline{\underline{291.78 \text{ Nm}}}$$

$$T = \mu F \frac{(r_i + r_o)}{2} \cdot n$$

$$F = 2\pi C(r_o - r_i)$$

$$F = 2\pi p_i r_i (r_o - r_i)$$

$$T = \mu 2\pi p_i r_i (r_o - r_i) \left(\frac{r_i + r_o}{2} \right) \cdot n$$

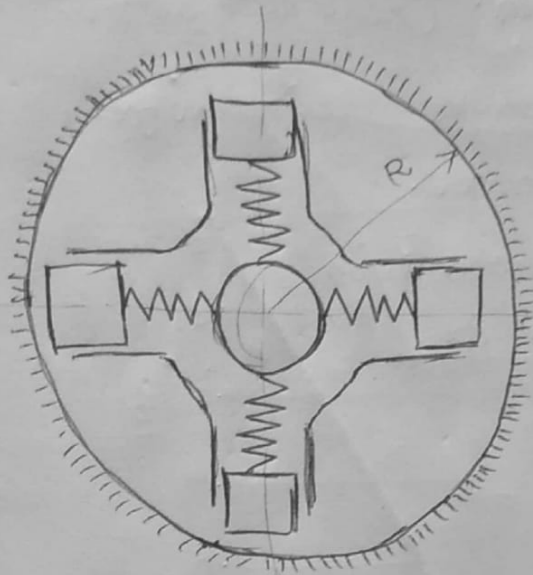
$$291.78 = 0.1 \times 2\pi \times 160 \times 10^3 \times 0.08 (0.106 - 0.08) \left(\frac{0.08 + 0.106}{2} \right) \cdot n$$

$$291.78 = 419.44 \times n$$

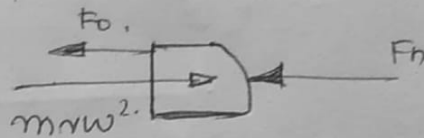
$$n =$$

Ans = 10.9
taken is 11

Centrifugal Clutches.



$$F_{c0} = m r \omega_0^2$$



$$m r \omega^2 - F_{c0}$$

$$F_h = m r \omega^2 - m r \omega_0^2$$

$$F = m r (\omega^2 - \omega_0^2) \mu R N$$

MODULE - CAMS

1. Draw the profile of a cam operating a knife edge follower, having a lift of 30 mm. The cam raises the follower with uniform velocity in 150° of rotation followed by a period of dwell for 60° . The follower descends for the next 100° rotation of the cam with uniform velocity. Again followed by a dwell period.

Cam rotates at uniform vel of 120 rpm,
and least radius of 20 mm.

What will be max. vel. and acc. of follower during lift and return.

A. $h = 30 \text{ mm.}$

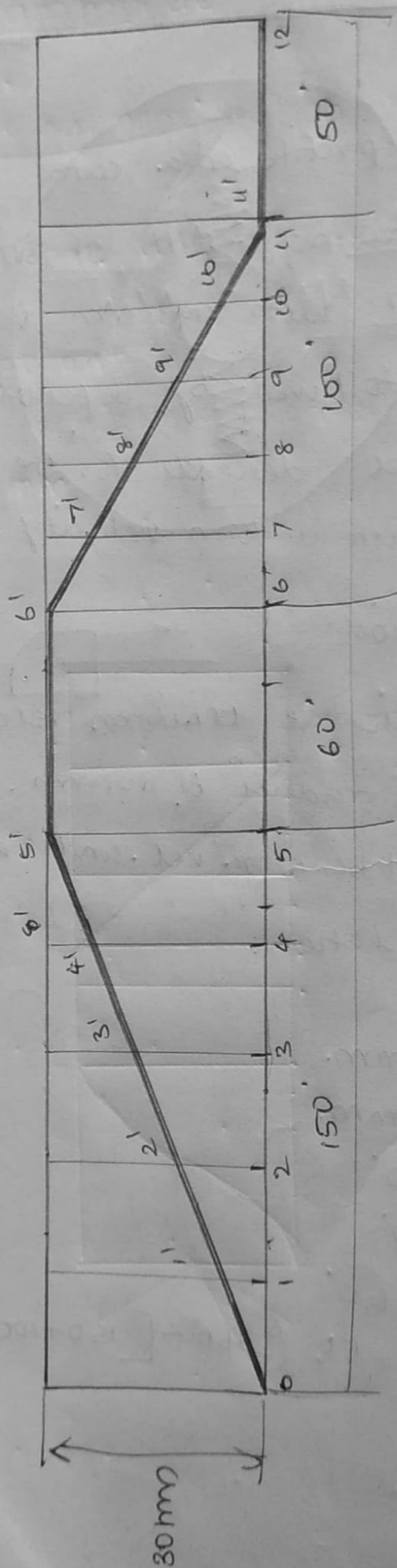
$r = 20 \text{ mm.}$

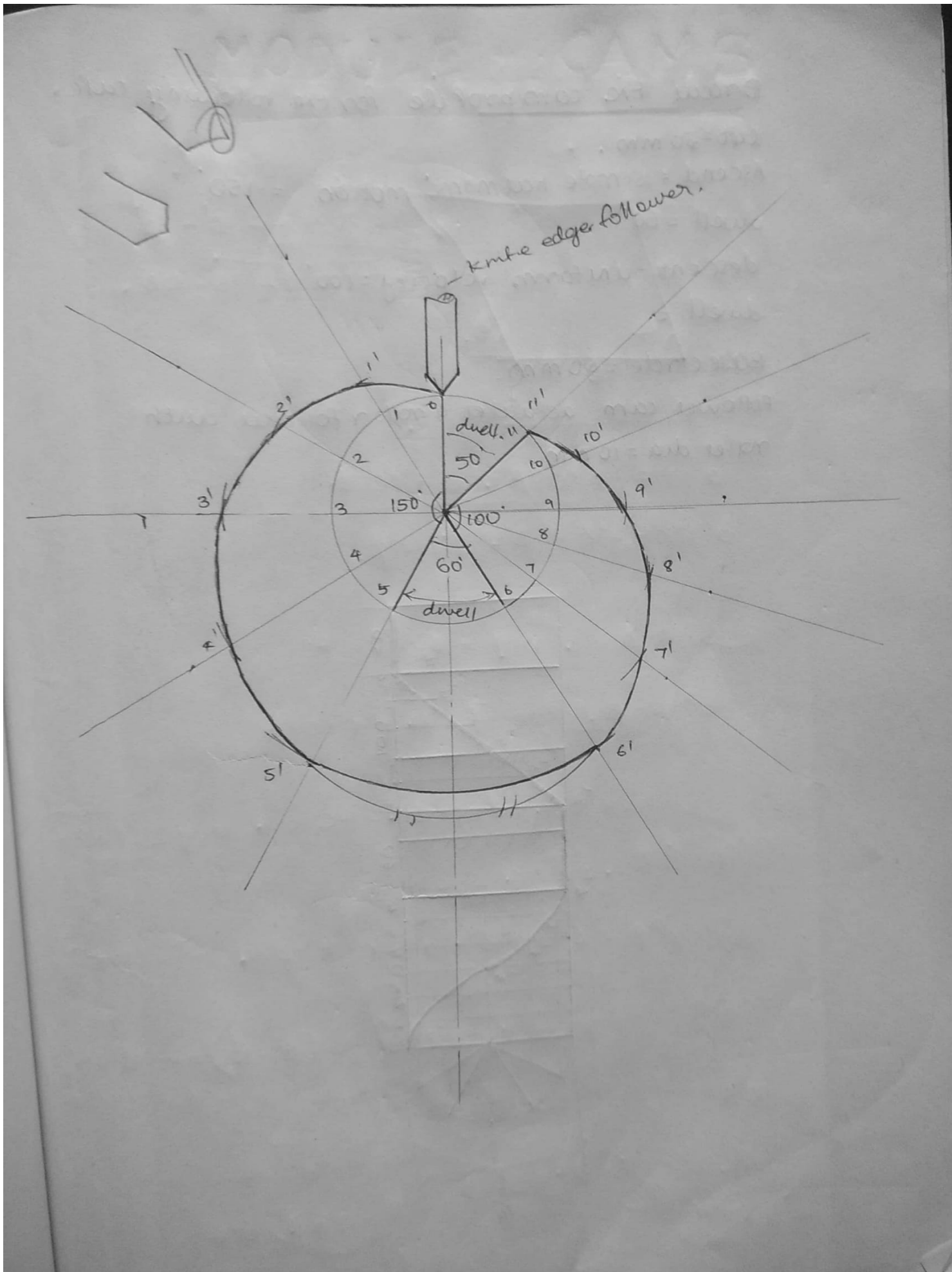
$\phi_a = 150^\circ$

$\phi_d = 100^\circ$

dwell = 60°

dwell = 50° $\left(360 - [150 + 100 + 60] \right)$





Draw the cam profile for the following case.

lift = 80 mm.

Ascend = simple harmonic motion = 150°

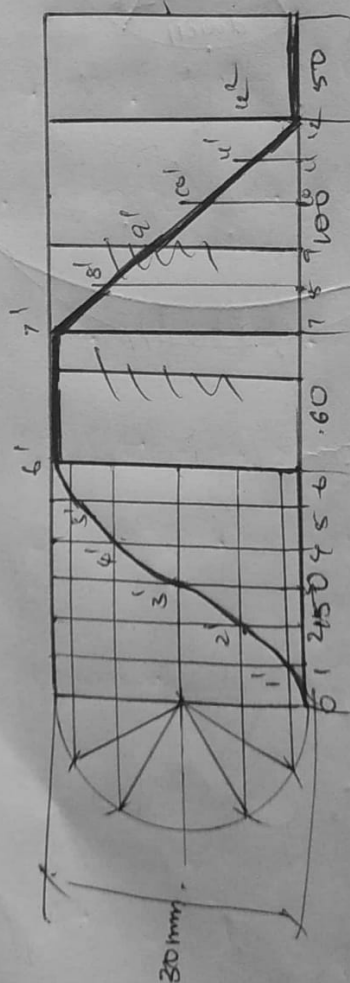
dwell = 60°

descent = uniform velocity = 100°

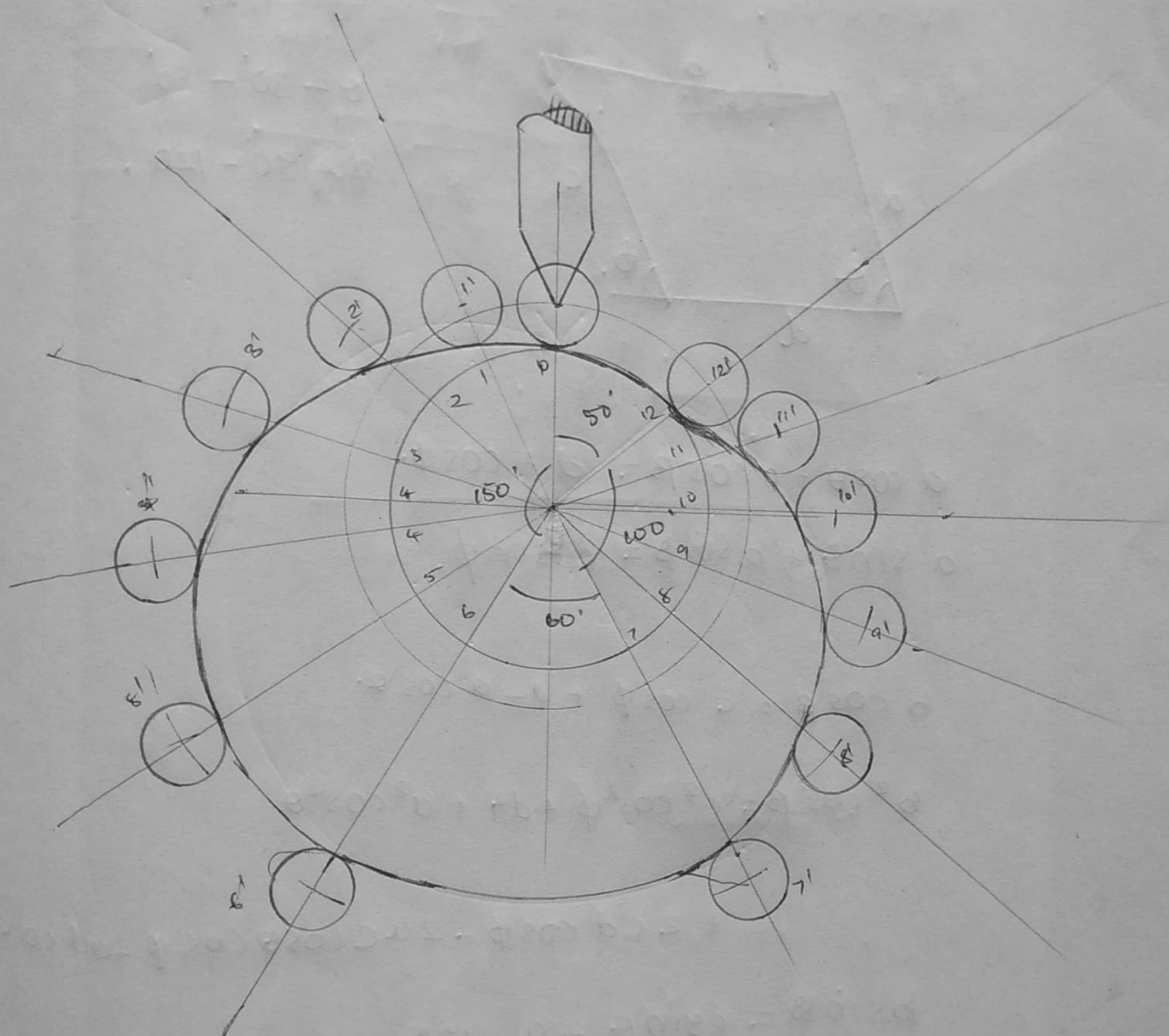
dwell =

Base circle = 20 mm.

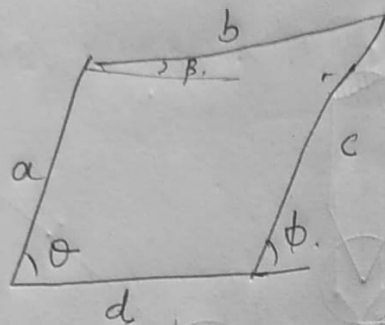
Follower cam actuates a roller follower with
roller dia = 10 mm.



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Synthesi's



$\theta \quad \phi$

$$y = f(x)$$

$$\theta = 30^\circ - 60^\circ$$

$$\phi = 90^\circ - 120^\circ$$

$$a \cos \theta + b \cos \beta = d + c \cos \phi$$

$$a \sin \theta + b \sin \beta = c \sin \phi$$

$$b \cos \beta = c \cos \phi + d - a \cos \theta$$

$$b^2 \cos^2 \beta = c^2 \cos^2 \phi + d^2 + a^2 \cos^2 \theta$$

$$+ 2cd \cos \phi - 2ac \cos \theta \cos \phi - ad \cos \theta$$

$$b \sin \beta = c \sin \phi - a \sin \theta$$

$$b^2 \sin^2 \beta = c^2 \sin^2 \phi + a^2 \sin^2 \theta - 2ac \sin \theta \sin \phi$$

$$b^2 = c^2 + a^2 + d^2 - 2ac \{ \cos(\theta - \phi) \} + 2cd \cos \phi - 2ad \cos \theta$$

or,

$$a^2 - b^2 + c^2 + d^2 + 2cd \cos \phi - 2ad \cos \theta = 2ac \cos(\theta - \phi)$$

Freudenstian eqn

$$\frac{d}{a} \cos \phi - \frac{d}{c} \cos \theta + \frac{a^2 - b^2 + c^2 + d^2}{2ac} = \cos(\theta - \phi)$$

$$k_1 \cos \phi - k_2 \cos \theta + k_3 = \cos(\phi - \theta)$$

1. Design a four-link mechanism to co-ordinate the 3 positions of the input and output links as follows.

$$\theta_1 = 20^\circ$$

$$\phi_1 = 35^\circ$$

$$\theta_2 = 35^\circ$$

$$\phi_2 = 45^\circ$$

$$\theta_3 = 50^\circ$$

$$\phi_3 = 60^\circ$$

Synthesize a mechanism.

A.

$$k_1 \cos 35^\circ - k_2 \cos 20^\circ + k_3 = \cos 15^\circ$$

$$k_1 \cos 45^\circ - k_2 \cos 35^\circ + k_3 = \cos 10^\circ$$

$$k_1 \cos 60^\circ - k_2 \cos 50^\circ + k_3 = \cos 10^\circ$$

$$k_1 (\cos 35^\circ + \cos 45^\circ)$$



$$k_1 = 0.643$$

$$d = 1$$

$$k_2 = 0.754$$

$$a = \frac{d}{k_1} = \frac{1}{0.643} = \underline{\underline{1.555}}$$

$$k_3 = 1.148$$

$$c = \frac{d}{k_2} = \frac{1}{0.754} = \underline{\underline{1.326}}$$

$$\frac{a^2 - b^2 + c^2 + d^2}{2ac} = k_3$$

$$b = \sqrt{a^2 + c^2 + d^2 - k_3 \times 2ac}$$

$$b = 1.248$$

Design a a four link mechanism when the motions of the input & output links are governed by a function $y = 2 \log_{10} x$ and x varies from 2 to 4

$$\theta = 30^\circ - 70^\circ$$

$$\phi = 40^\circ - 100^\circ$$

	x	θ	y	ϕ
x_1	2	30	0.602	40
x_2	3	50	0.954	75
x_3	4	70	1.204	100

$$\theta_i = \theta_1 + \frac{(x_i - x_F)}{(x_F - x_S)} (\theta_F - \theta_S)$$

θ_S - starting
 θ_F - final

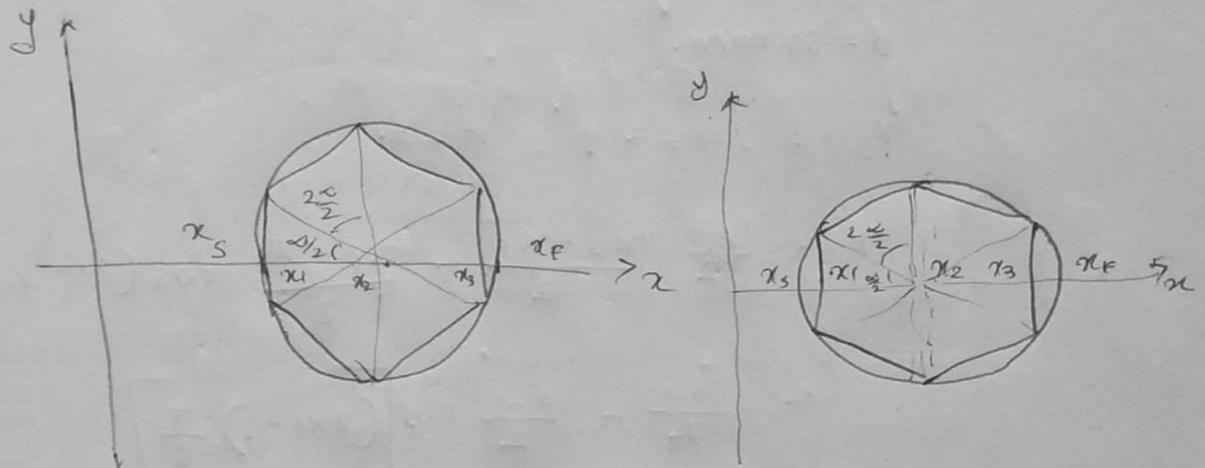
$$\phi_2 = 40 + \frac{(3-2)(100-40)}{(4-2)}$$

$$\phi_2 = 40 + \frac{(0.954-0.602)}{(1.204-0.602)}$$

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$$k_1 \cos \phi + k_2 \cos \theta + k_3 = \cos (\phi - \theta)$$

chebyshev spacing



$$x_i = \frac{x_s + x_F}{2} - \frac{(x_F - x_s)}{2} \cos \left((2i-1) \frac{\pi}{2n} \right)$$

- 1). Design a fourline mechanism if the motions of the input and output links are governed by a function $y = x^{1.5}$ and x varies from 1 to 4. Assume θ to vary from 30 to 120 and ϕ from 60 to 130. The length of fixed link is 30 mm. Use chebyshev's spacing accuracy points

A Given.

$$y = x^{1.5}$$

$$x = 1 \text{ to } 4$$

$$\theta \rightarrow 30^\circ - 120^\circ$$

$$\phi \rightarrow 60^\circ - 130^\circ$$

$$d = 30 \text{ mm.}$$

$$x_S = 1, x_F = 4, n = 3.$$

$$x_i = \frac{x_S + x_F}{2} - \frac{x_F - x_S}{2} \cos \left((2i-1) \frac{\pi}{2n} \right)$$

$$= \frac{1+4}{2} - \frac{4-1}{2} \cos \left[(2 \times 1 - 1) \frac{\pi}{6} \right]$$

$$= \frac{1.2009}{1.00006}$$

$$x_2 = 2.5 - 1.5 \cos 3 \times \frac{\pi}{6} = 2.5$$

$$x_3 = 2.5 - 1.5 \cos 5 \times \frac{\pi}{6} = 3.799$$

	x	θ	$y = x^{1.5}$	ϕ
S	1	30	1	60
1	1.201	36	1.316	63.2
2	2.5	75	3.952	89.5
3	3.799	114	7.404	123.7
F	4	120	8	130

$$\theta_i = 30 + \frac{(4-1)}{(120-30)} (1.201-1)$$

$$\theta_i = \theta_s + \frac{(x_i - x_s)}{(\theta_F - \theta_s)} (\theta_F - \theta_s)$$

$$\theta_i = \theta_s + \frac{(x_i - x_s)}{(\theta_F - \theta_s)} (\theta_F - \theta_s)$$

$$\theta_1 = 30 + \frac{(120-30)}{(4-1)} (1.201-1)$$

$$= \underline{\underline{36.03}}$$

$$\theta_2 = 30 + \frac{90}{3} \times (2.5-1) = \underline{\underline{75}}$$

$$\theta_3 = 30 + \frac{90}{3} (3.799-1) = \underline{\underline{114}}$$

$$\phi_i = \phi_s + \frac{y_i - y_s}{y_F - y_s} (\phi_F - \phi_s)$$

$$\phi_1 = 60 + \frac{70}{7} \times (1.316-1) = 63.2$$

$$\phi_2 = 60 + \frac{70}{7} \times (3.952-1) = 89.5$$

$$\phi_3 = 60 + \frac{70}{7} \times (7.404-1) = 123.7$$

CAM

- 1) Uniform velocity diagram
- 2) SHM
- 3) Uniform acceleration & deceleration
- 4) Cycloidal motion.

1) . Draw the profile of a cam operating a roller reciprocating follower with the following data

Minimum radius of cam = 25 mm

Lift = 30 mm.

Roller dia = 15 mm.

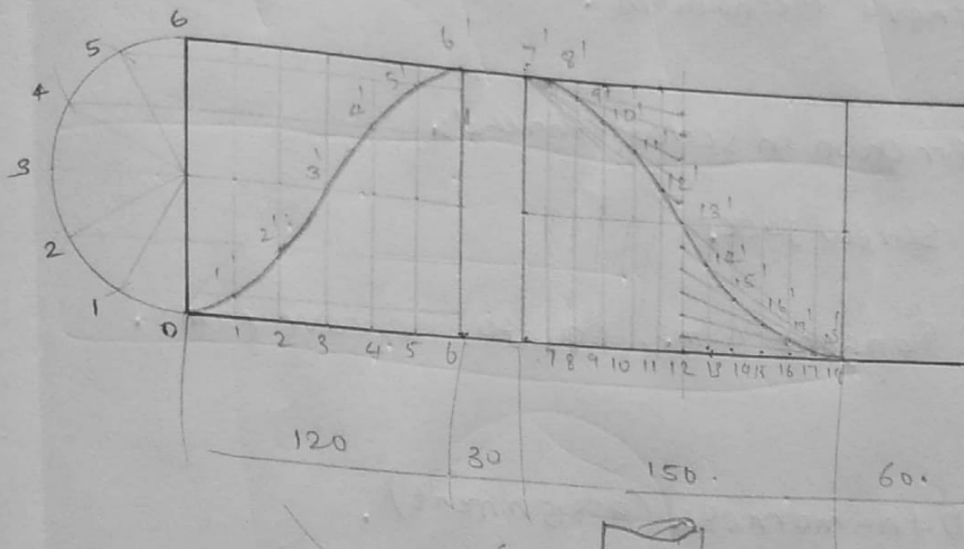
Cam lifts the follower for 120° with SHM, followed by a dwell period of 30° .

Then follower lowers down during 150° of cam rotation with uniform acc & deceleration followed by a dwell period.

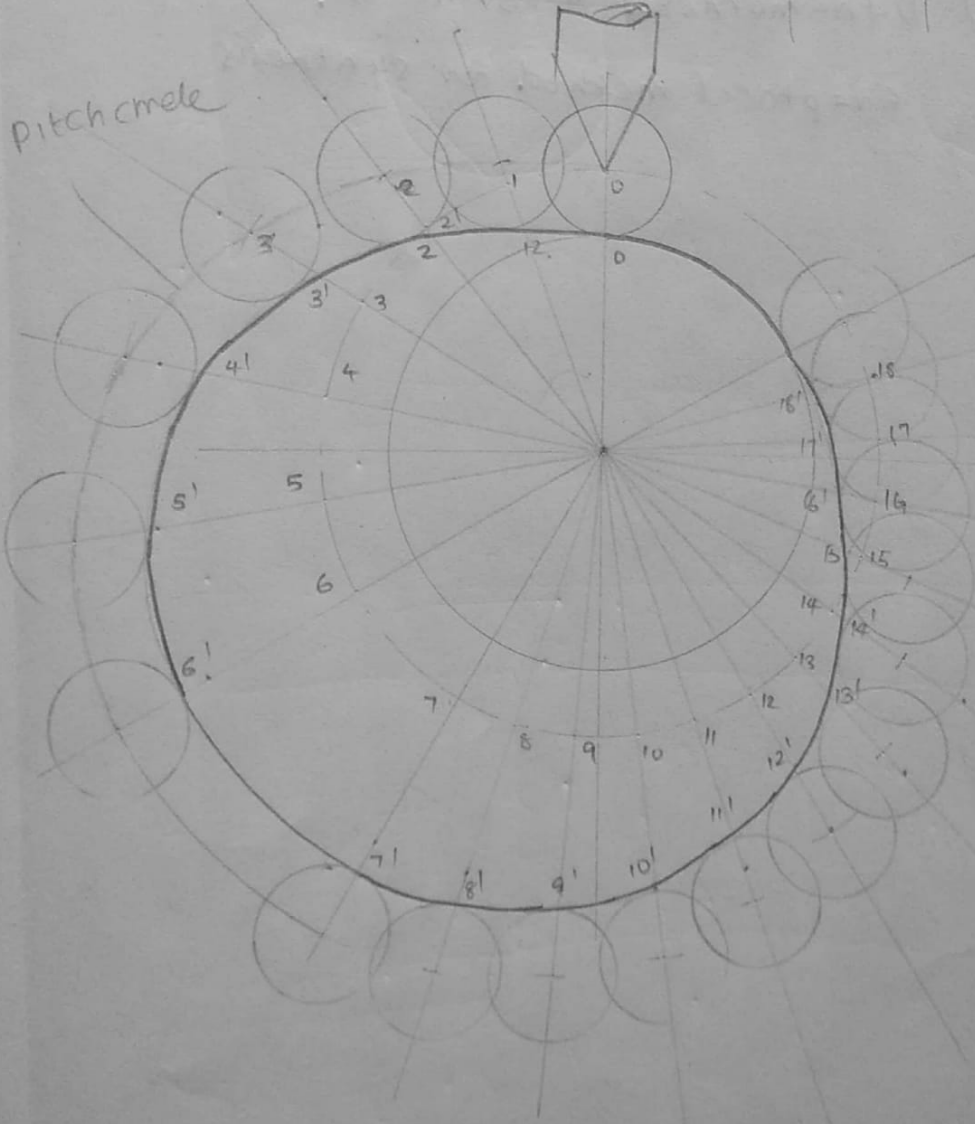
If the cam rotates at uniform speed of 150 rpm in clockwise direction calculate the max. velocity and acceleration of follower during descending period.

A.

Displacement Diagram



Pitch circle



cam profile

Draw the cam profile for the previous problem
if the follower has an offset of 10 mm

